Noun Rank Benchmarking System v6.0

AI LLM Accuracy Validation for Long Document Summarization

GovReport-Calibrated Validation Algorithm

1. Overview

This document presents a benchmark-calibrated algorithm for validating AI-generated summaries using noun-based analysis. The system uses the GovReport dataset to establish baseline scores from expert-written summaries, then measures LLaMA 70B accuracy against these calibrated baselines.

1.1 Problem Statement

How to objectively measure the accuracy of AI LLM systems for Long Document Summarization using calibrated benchmarks and expert-validated ground truth.

1.2 Key Innovation

This version introduces a two-stage calibration approach:

1. **Calibration Stage:** Establish baseline scores using expert-written summaries from GovReport
2. **Evaluation Stage:** Measure LLaMA 70B accuracy as a percentage of expert baseline

This approach replaces the previous dual-model comparison (LLaMA vs Mistral) with a more rigorous expert-calibrated methodology.

1.3 System Specifications

Parameter	Value
Target Model	LLaMA 70B (INT8 quantization)
Validation Method	Noun Rank Algorithm (calibrated)
Benchmark Dataset	GovReport (19,466 documents)
Ground Truth	Expert-written summaries
Input	1-50 page documents
Output	1-5 page document summary + accuracy score

1.4 What Changed from v5

Component	v5 (Previous)	v6 (Current)
MMLU Comparison	Not addressed	New Section 7 added
Benchmark Justification	Implicit	Explicit MMLU limitations documented

2. Calibration Methodology

The system operates in two distinct stages to ensure rigorous validation against expert ground truth.

2.1 Stage 1: Baseline Calibration (One-Time Setup)

This stage establishes what a "perfect" summary scores by running Noun Rank on expert-written summaries from GovReport.

Process:

1. Load GovReport dataset (19,466 document-summary pairs)
2. For each document, run Noun Rank algorithm comparing source → expert summary
3. Store baseline score for each document in Reference Score Database
4. Calculate dataset statistics (mean, std, percentiles)

2.2 Stage 2: LLaMA Evaluation (Ongoing)

This stage measures LLaMA 70B accuracy against the calibrated baselines.

Process:

1. Load source document from GovReport
2. Generate summary using LLaMA 70B
3. Run Noun Rank algorithm comparing source → LLaMA summary
4. Retrieve baseline score for this document
5. Calculate accuracy: LLaMA Score ÷ Baseline Score × 100%

Accuracy Formula:

Accuracy = (LLaMA Noun Rank Score ÷ Expert Baseline Score) × 100%

Interpretation:

An accuracy of 92% means "LLaMA's summary is 92% as good as a human expert's summary" — a meaningful, objective measure.

3. Scoring Components

The Noun Rank algorithm uses five weighted components to calculate a comprehensive score.

Component	Symbol	Weight	Purpose
Hallucination Score	H	0.25	Penalize fabricated content not in source
Coverage Score	C	0.25	Reward inclusion of key facts from source
Frequency Score	F	0.20	Weight nouns by source importance
Entity Score	E	0.20	Validate named entities strictly
Semantic Score	S	0.10	Overall meaning alignment

Final Score Formula:

Score = $(1 - H) \times 0.25 + C \times 0.25 + F \times 0.20 + E \times 0.20 + S \times 0.10$

4. Algorithm Steps

The algorithm follows a seven-step process: (1) Extraction & Normalization, (2) Hallucination Score calculation, (3) Coverage Score calculation, (4) Frequency Score calculation, (5) Entity Score calculation, (6) Semantic Score calculation, and (7) Final Score & Accuracy Calculation. Refer to the detailed technical appendix for complete formulas.

5. Accuracy Interpretation

The accuracy score represents how well LLaMA's summary compares to an expert-written summary for the same document.

Accuracy	Level	Interpretation
95-100%	Excellent	Near expert quality, production ready
85-94%	Good	High quality, acceptable for most use cases

70-84%	Acceptable	Moderate quality, may need review
50-69%	Low	Significant issues, requires manual review
<50%	Reject	Unacceptable, do not use

6. GovReport Benchmark Dataset

GovReport serves as both the calibration source and evaluation benchmark for the Noun Rank system.

6.1 Dataset Overview

Feature	Value
Total Documents	19,466 document-summary pairs
Average Document Length	~9,400 words
Average Summary Length	~550 words
Summary Authors	Government experts (CRS & GAO)
Train/Val/Test Split	17,517 / 974 / 975
Access	Hugging Face: ccdv/govreport-summarization

6.2 Why GovReport is Ideal

- **Expert-Written Summaries —** Ground truth written by the same experts who authored the reports
- **Long Documents —** Average 9,400 words matches target use case of 1-50 page documents
- **Factual Content —** Government reports contain verifiable facts, entities, and data points
- **Formal Structure —** Professional documents with clear sections for reliable NLP processing
- **Free & Accessible —** Available on Hugging Face with no licensing restrictions

7. Limitations of Traditional Benchmarks: Why MMLU Falls Short

The Massive Multitask Language Understanding (MMLU) benchmark has become a widely cited metric for evaluating LLM capabilities. However, MMLU has fundamental limitations that make it inadequate for measuring real-world performance in tasks like document summarization. This section outlines why the Noun Rank Algorithm addresses gaps that MMLU cannot.

7.1 Static Dataset and Training Bias

- **Stale Benchmark —** MMLU has been a static dataset since its release in 2020. The same 15,908 questions have been used for years without updates, increasing the risk that LLM training data has been inadvertently (or deliberately) optimized for MMLU-style questions.
- **Data Contamination Risk —** As MMLU questions circulate widely in public datasets, newer LLMs may have seen these exact questions during training, artificially inflating benchmark scores without reflecting genuine capability improvements.
- **Benchmark Gaming —** LLM developers may bias implementations toward MMLU-favored patterns, question structures, and answer formats, resulting in models that perform well on the benchmark but underperform on novel, real-world tasks.

7.2 Multiple Choice Does Not Reflect True Comprehension

- **Recognition vs. Generation —** MMLU tests the ability to recognize the correct answer among four options. This is fundamentally different from generating accurate, coherent, and complete content—the actual requirement for summarization tasks.
- **No Measure of Reasoning Depth —** A correct multiple-choice answer does not prove the model understood the underlying concepts. Models can exploit statistical patterns, eliminate obviously wrong answers, or guess based on surface-level cues without true comprehension.
- **Binary Scoring Limitation —** MMLU provides only right/wrong feedback. There is no nuance for partially correct reasoning, no credit for valid alternative interpretations, and no penalty for confident wrong answers versus uncertain guesses.
- **No Hallucination Detection —** MMLU cannot detect whether a model fabricates information. In summarization, hallucination detection is critical—a model might generate fluent, confident text that contains entirely false claims.

7.3 Focus on Training Data Extensiveness, Not Task Performance

- **Knowledge Recall Bias —** MMLU primarily measures how much factual knowledge a model has memorized across 57 subjects. High scores indicate extensive training data coverage, not necessarily the ability to apply knowledge to practical tasks.
- **No Task-Specific Evaluation —** MMLU does not evaluate summarization, document analysis, report generation, or other practical NLP applications. A model scoring 90% on MMLU may still produce poor summaries.
- **Compute Power Conflation —** Increased compute power primarily accelerates inference speed and enables larger batch processing. MMLU scores may improve with scale, but this reflects training investment, not improved comprehension or reasoning quality for real-world tasks.

7.4 What Noun Rank Offers That MMLU Cannot

Capability	MMLU	Noun Rank
Hallucination Detection	✗ Not measured	✓ Core component (25% weight)
Factual Coverage	✗ Not measured	✓ Key noun & entity tracking
Long Document Processing	✗ Short Q&A format	✓ Designed for 1-50 pages
Generation Quality	✗ Tests recognition only	✓ Evaluates generated output
Expert Calibration	✗ No ground truth baseline	✓ Calibrated to expert summaries
Nuanced Scoring	✗ Binary right/wrong	✓ Multi-component weighted score
Named Entity Validation	✗ Not measured	✓ Strict entity matching (20%)
Dynamic Evaluation	✗ Static since 2020	✓ Any document can be evaluated

7.5 Summary

MMLU serves a purpose as a broad measure of factual knowledge breadth, but it fundamentally cannot evaluate what matters for document summarization: the ability to extract key information, avoid fabrication, maintain factual accuracy, and produce coherent output that matches expert quality. The Noun Rank Algorithm directly addresses these gaps by providing task-specific, hallucination-aware, expert-calibrated evaluation that reflects real-world summarization performance.

8. System Architecture

This section defines the recommended technology stack and infrastructure for implementing the Noun Rank Benchmarking System.

8.1 Building Blocks

Component	Technology
Target Model	LLaMA 70B (INT8 quantization)
Validation Method	Noun Rank Algorithm
LLM Inference	Vast.ai (recommended)
Benchmark Dataset	GovReport (19,466 documents)
NLP Library	spaCy (noun extraction, NER, lemmatization)
Embeddings	sentence-transformers
Programming Language	Python

9. Conclusion

Version 6.0 of the Noun Rank system builds upon the expert-calibrated methodology introduced in v5, with the addition of a comprehensive analysis of why traditional benchmarks like MMLU are insufficient for evaluating summarization quality.

Key Innovations:

- **Expert-Calibrated Baselines —** Using GovReport's expert summaries as ground truth
- **Absolute Accuracy Measurement —** LLaMA accuracy as percentage of expert quality
- **MMLU Limitations Documented —** Clear justification for why Noun Rank provides superior evaluation
- **Hallucination-Aware Scoring —** Direct measurement of fabricated content, which MMLU cannot detect
- **Task-Specific Evaluation —** Purpose-built for document summarization rather than generic Q&A

This approach provides a meaningful, objective, and actionable metric: when the system reports "92% accuracy," it means LLaMA's summary is 92% as good as an expert-written summary—a claim that MMLU scores fundamentally cannot support.